

Algebra 1 - Q2.2_1.3 A

Name: _____

2.1 – Expressions and Their Parts

1. I can identify the parts of an algebraic expression

In the following expression, list any **terms**, **coefficients**, and **constants**.

$$3x^2 - 2x + 5$$

Terms:

Coefficients:

Constants:

2. I can describe what a variable, constant, coefficient, and term are

Answer each question about the expression $5x - 2y + 3x^2 - 9$ with **true** or **false**.

- a. The expression has 3 terms _____
- b. The coefficients are 5, -2, and 3 _____
- c. The constant is 9 _____
- d. Both x and y are variables in the expression _____

3. I can write a simple expression to model a real-world situation

Your boss says she will pay you \$50 to show up to work plus \$12 for every hour you work. Write an expression that represents your total pay. Let h be the number of hours you work.

2.2 – Evaluating Expressions

4. I can substitute values into expressions and simplify them

Evaluate the following expressions.

a. $5x - 7$ when $x = -2$

b. $-a^2 + 3b - ab$ when $a = -4$ and $b = 2$

5. I can use the correct order of operations with variables and parentheses

Evaluate the following expressions.

a. $2(x - 1)^2 + 5x$ when $x = 2$

b. $3|2y - x| + 5y$ when $x = 8$ and $y = -10$

c. $\frac{1}{x}(2x - 3)^2 + x$ when $x = 3$

6. I can use substitution to test whether two expressions are equivalent

Test whether $3x - 9$ is equivalent to $3(x - 2) - 3$. Try **2 different values** for x .

2.3 – Combining Like Terms

7. I can identify like terms in an expression

For each of the following, indicate whether the terms are like terms (**true** or **false**).

a. $5x$ and $-5x$ _____

b. $-x^2$ and $-x^3$ _____

c. $2xy$ and $3yx$ _____

d. $5x^2y$ and $-9xy^2$ _____

8. I can combine like terms to simplify an expression

Simplify the following expressions by combining like terms.

a. $52 - 3x^2 + 5x - 11x + 3$

b. $a + 2b - 3a + b^2 - 5 + 2b^2 - 1$

9. I can check my work by substitution or by reasoning about equivalence

Simplify the following expression

$$5x - 3y^2 + 8x - 2 - y$$

Test whether your answer is equivalent to the original expression.